

CCP 2: Urdu Text Generation Using Deep Learning Models

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Submission Deadline

[30 June, 2025]

Objective

The goal of this assignment is to implement deep learning models for generating coherent Urdu text. Students will use a provided dataset of 18,814 Urdu sentences to train and evaluate models such as RNN, LSTM, GRU, or Transformer-based architectures.

Dataset

The corpus consists of **18,814 Urdu sentences**, with each sentence occupying a separate line. These will be used to train your text generation model.

Tasks

1. Data Preprocessing

- Tokenize the Urdu text (either word-level or character-level).
- Create input-output pairs for sequence modeling.
- Perform padding or truncation as needed.
- Encode the text using one-hot encoding or an embedding layer.

2. Model Implementation

Implement following models for comparative analysis:

- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Transformer

- Fine tuning of pretrained models

3. Training and Optimization

- Use a suitable loss function (e.g., categorical crossentropy).
- Use an appropriate optimizer (e.g., Adam).
- Train the model on the sequence data.

4. Text Generation

- Generate Urdu text based on seed input.
- Use temperature sampling or top-k sampling for diversity.
- Showcase 5–10 generated sentences.

5. Evaluation and Analysis

- Train all models and plot training loss for all models.
- Generate Urdu sentences from the same seed
- Compare output length, and quality in terms of (grammar, fluency, and coherence).
- Comment on challenges such as tokenization, vocabulary size, model convergence.

Deliverables

- Python notebook or script (.ipynb or .py)
- A short report summarizing methodology, results, and analysis (PDF/Word)
- Sample generated Urdu text
- Trained model weights (.h5 or .pt)

Important Notes

- Ensure all Urdu text is handled using UTF-8 encoding.
- Make sure text is properly rendered right-to-left (RTL).
- Reduce sequence length and vocabulary size if training time is too long.

Recommended Libraries

- TensorFlow / Keras or PyTorch
- HuggingFace Transformers (optional)
- nltk or hazm for preprocessing
- matplotlib, seaborn for plotting